Notes on Distributions, Sobolev Spaces and Weak Solutions

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The main objective of this text is to compile everything I've read or discussed about weak solutions on PDEs and the prerequisites for understanding this topic. The reader may notice that some of the demonstrations of the early parts of this text are omitted, as they are not the main focus of this text. But for the interested, they all can be found in the book "Topics in functional analysis and applications" listed in the bibliography,

Summary

CHAPTER 1

Motivation

We will begin by studying the solution of the following equation:

$$u_t + uu_x = 0, \text{ for } x \in \mathbb{R} \text{ and } t > 0 \tag{1}$$

This is commonly known as the inviscid Burger's equation, a simplified model for fluid mechanics, modelling one dimensional flow where $u(x,t)$ is the velocity of the particle located in x at time t. So, first, we are going to try to find regular solutions for this equation by the method of characteristics.

Considering $u(x,t)$ to be a smooth solution for (1) satisfying the initial condition problem:

$$u(x,0) = u_0(x)$$

Where $u_0(x)$ is a function of x. We'll start by defining the characteristic curves:

$$\begin{cases} \dot{x}(t) = u(x(t),t),\ x(0) = x_0 \\ \dot{y}(t) = 1,\ y(0) = y_0 \end{cases}$$

Solely from that, we get $y = t$. And now, by calculating the total derivative of u on the curves just defined, we obtain:

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{\partial u}{\partial t}\frac{\partial y}{\partial t} + \frac{\partial u}{\partial x}\frac{\partial x}{\partial t} = u_t + u_x u = 0$$

From this, two conclusions are made: u is constant along the curve $(x(t),t)$, which implies that $\dot{x}(t) = u_0(x_0)$. By solving this last ODE, we conclude that:

$$x = x_0 + u_0(x_0)t$$

And for all points in this curve $u(x,t) = u_0(x_0)$. For convenience's sake, we will choose this as our initial curve u_0:

Using the initial condition given by u_0, the characteristic curves are:

We can clearly see that there are curves intersecting each other, so the value of $u(x,t)$ is not well-defined at the intersection points, therefore this cannot be a solution for our equation.

Now, the question that rises is: "if that isn't a solution, then what is?". And that will motivate our next few chapters of functional analysis definitions and theorems, aiming to build a generalized perspective on what is a solution for a PDE.

CHAPTER 2

Test functions and Distributions

Our main objective in this chapter is to make useful definitions to generalize the derivative of a function. For that, we will need a mechanism that can weaken the conditions for a function to have partial derivatives, by removing it entirely and passing it to another convenient function that always has a derivative in any axis.

2.1 The elements of $\mathscr{D}(\Omega)$ and $\mathscr{D}'(\Omega)$

Definition 2.1: Test functions

Let Ω be a subset of $\mathbb{R}^n$. Then $\mathscr{D}(\Omega)$ is the set of all C^∞ functions defined on Ω with compact support.

An example of such function is:

$$f(x) = \begin{cases} \exp(\frac{-a^2}{a^2-x^2}), |x| < a \\ 0, |x| \geq a \end{cases}$$

And, for any φ on this set and any locally integrable function $f : \mathbb{R}^n \to \mathbb{R}$, using integration by parts we get:

$$\int_K D^\alpha f \varphi = (-1)^{|\alpha|} \int_K f D^\alpha \varphi \tag{2}$$

Where $K \supseteq \text{supp}\varphi$ is a compact subset of Ω and α is a multi-index. This will motivate our next definition, but before that, we need to quickly describe the topology of $\mathscr{D}(\Omega)$:

A sequence $\{\varphi_n\}_{n\in\mathbb{N}}$ in $\mathscr{D}(\Omega)$ converges to φ if there exists a compact set $K \subset \Omega$ such that $\text{supp}(\varphi_n) \in K$ for all n and $D^\alpha\varphi_n \to D^\alpha\varphi$ for all α.

Now we can define the most important tool used in this endeavour:

Definition 2.2: Distributions

A linear functional T is said to be a distribution if for all sequences $\{\varphi_n\}_{n\in\mathbb{N}}$ in $\mathscr{D}(\Omega)$ convergent to some φ, the sequence $\{T\varphi_n\}_{n\in\mathbb{N}}$ converges to $T\varphi$ in $\mathscr{D}(\Omega)$.

We will use $\mathscr{D}'(\Omega)$ for the set of all distributions. Note that $\mathscr{D}'(\Omega)$ uses the same notation as the dual space of $\mathscr{D}(\Omega)$ and that is no coincidence, as the distributions are precisely the dual elements of the test functions, as they are the linear functionals defined from $\mathscr{D}(\Omega)$ to $\mathbb{R}$. Furthermore, it is worth to point that both $\mathscr{D}(\Omega)$ and $\mathscr{D}'(\Omega)$ are vector spaces over $\mathbb{R}$.

Now, aiming for (2), we will define the following:

Definition 2.3: Distribution given by a function

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a locally integrable function, we say that the distribution given by f is:

$$T_f(\varphi) = \int_\Omega f\varphi$$

It is trivial to verify that T_f is in fact a distribution, because as $\varphi_n \to \varphi$, $\int_\Omega f\varphi_n \to \int_\Omega f\varphi$.

Another useful distribution is the Dirac "δ-function", which is in fact a distribution given by:

Definition 2.4: The Dirac distribution

For any $\varphi \in \mathscr{D}(\Omega)$:

$$\delta(\varphi) = \varphi(0)$$

Both distributions will be essential for the next sections.

Lastly, we'll define the support of a distribution:

Definition 2.5: T

e support of a distribution T is the complement of the largest set in Ω in which for all test functions $T(\varphi) = 0$

By this definition we can conclude that, if $T = T_f$ for some locally integrable function, then:

$$\text{supp}(T_f) = \text{supp}(f)$$

And for the Dirac δ distribution (also for all of its derivatives that will be defined):

$$\text{supp}(\delta) = \{0\}$$

2.1.1 Multiplication of a distribution by a function

From now on, we will use the notation $\mathscr{E}(\Omega)$ for the C^∞ functions defined in Ω. First, let $\{\varphi_n\}_{n\in\mathbb{N}}$ be a sequence of functions in $\mathscr{D}(\Omega)$ converging to φ, notice that, for $\psi \in \mathscr{E}(\Omega)$, $\{\psi\varphi_n\}_{n\in\mathbb{N}}$ converges to φ. Additionally, as φ_n has compact support, $\psi\varphi$ will also. Therefore, if T is a distribution, the sequence $\{T(\psi\varphi_n)\}_{n\in\mathbb{N}}$ is convergent to $T(\psi\varphi)$. So:

$$\psi T(\varphi) = T(\psi\varphi)$$

defines a distribution. If T is a distribution given by a function f, we get that:

$$\psi T_f(\varphi) = T_f(\psi\varphi) = \int_\Omega f\psi\varphi = T_{f\psi}(\varphi)$$

2.2 Distributional derivatives

Having (2) in mind, let's build a definition of a distribution derivative that can satisfy that equation. Let f be a smooth function in Ω, so:

$$D^\alpha Tf(\varphi) = D^\alpha \int_\Omega f\varphi$$

Now, by the dominated convergence theorem we can push the derivative inside the integral, that is:

$$D^\alpha \int_\Omega f\varphi = \int_\Omega D^\alpha f\varphi = (-1)^{|\alpha|} \int_K fD^\alpha\varphi$$

So, we now know what we have to define $D^\alpha T$ as to serve our needs. We let f as a smooth function just so $D^\alpha f$ has some meaning in a sense, but in general, the only condition f must abide is to be locally integrable.

Definition 2.6: Derivative of a distribution

Let $T \in \mathscr{D}'(\Omega)$, the α derivative of T is:

$$D^\alpha T(\varphi) = (-1)^{|\alpha|} T(D^\alpha\varphi)$$

We need to verify that $D^\alpha T$ is also a distribution in $\mathscr{D}'(\Omega)$:

Theorem 2.1

For $T \in \mathscr{D}'(\Omega)$ and any multi-index α, $D^\alpha T$ is also in $\mathscr{D}'(\Omega)$.

Proof. Let $\{\varphi_n\}_{n\in\mathbb{N}}$ be a sequence of functions in $\mathscr{D}(\Omega)$ converging to φ, then by the topology of $\mathscr{D}(\Omega)$, $\{D^\alpha\varphi_n\}_{n\in\mathbb{N}}$ converges to $D^\alpha\varphi$. Additionally, have that $\{T(\varphi_n)\}_{n\in\mathbb{N}}$ converges to $T\varphi$, therefore, by definition 2.6:

$$D^\alpha T(\varphi_n) = (-1)^{|\alpha|} T(D^\alpha\varphi_n)$$

Because $\{D^\alpha\varphi\}_{n\in\mathbb{N}}$ is a converging sequence, $\{T(D^\alpha\varphi_n)\}_{n\in\mathbb{N}}$ converges to $T(D^\alpha\varphi)$ and then $D^\alpha T \in \mathscr{D}'(\Omega)$.

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Another important thing to note is that, for a distribution given by a function f, the following identity is verified:

Lemma 2.1

$$D^\alpha T_f(\varphi) = T_{D^\alpha f} = (-1)T_f(D^\alpha \varphi)$$

Which was already proven before. In addition to that, we will say that a locally integrable function f has a distributional derivative if there exists a distribution G such that for all $\varphi \in \mathscr{D}(\Omega)$:

$$\frac{\mathrm{d}}{\mathrm{d}t} T_f(\varphi) = G(\varphi)$$

So G is the distributional derivative of f. If $G = T_g$ for some locally integrable function g, then we also say that g is the distributional derivative of f. This distributional derivative is also what is called the weak derivative of f, sometimes denoted as $\partial_t f$.

It should be clear that if a function has a regular derivative, its weak derivative will be the same, as the identity above will be satisfied by:

$$\frac{\mathrm{d}}{\mathrm{d}t} T_f = T_{\frac{\mathrm{d}f}{\mathrm{d}t}}$$

Example 2.1: The derivative of the Heaviside Function

The Heaviside function is defined as:

$$H(x) \begin{cases} 1, x \geq 0 \\ 0, x < 0 \end{cases}$$

So the distributional derivative of H is the δ distribuition.

Proof. Let $\Omega \in \mathbb{R}$. As H is a locally integrable function, let's evaluate the distribution given by H:

$$T_H(\varphi) = \int_\Omega H(x)\varphi(x)dx$$

So, by taking the derivative of this distribution and using lemma 2.1 we get:

$$\frac{\mathrm{d}}{\mathrm{d}x} T_H(\varphi) = -T_H(\frac{\mathrm{d}\varphi}{\mathrm{d}x}) = -\int_0^{+\infty} H(x)\frac{\mathrm{d}\varphi}{\mathrm{d}x}dx = \varphi(x)\Big|_0^{+\infty} = \varphi(0) = \delta(\varphi)$$

Therefore, δ is the distributional derivative of the Heaviside function. ∎

Example 2.2: The derivatives of the δ distribution

For $\Omega \subseteq \mathbb{R}$ and $\varphi \in \mathscr{D}(\Omega)$, using definition 2.6, we get:

$$\delta^{(n)}(\varphi(x)) = (-1)^n \delta(\varphi^{(n)}(x)) = (-1)^n \varphi^{(n)}(x)$$

Where $\varphi^{(n)}$ is the n-nht derivative of φ.

We'll finish this section by giving some useful properties for the derivative of a distribution:

Proposition 2.1: The derivative of a distribuition multiplied by a $\mathscr{E}(\Omega)$ function

By definition 2.6:

$$D^\alpha(\psi T)(\varphi) = D^\alpha T(\psi\varphi) = (-1)^{|\alpha|}T(D^\alpha\psi\varphi) = (-1)^{|\alpha|}(D^\alpha\psi)T(\varphi)$$

Theorem 2.2: Leibniz formula for distributions

For $\psi \in \mathscr{E}(\Omega)$, $T \in \mathscr{D}'(\Omega)$ and α a multi-index:

$$D^\alpha(\psi T) = \sum_{\beta\leq\alpha} \frac{\alpha!}{\beta!(\alpha-\beta)!} D^\beta\psi D^{\alpha-\beta}T$$

2.3 Convolution of functions and distributions

Let's first remind ourselves of the definition of a convolution between two functions. For f,$g \in L_1(\mathbb{R}^n)$, the convolution of f and g is defined as:

$$f * g(x) = \int_{\mathbb{R}^n} f(x-y)g(y)dy$$

It is important also to point out that this operation is commutative and associative as to keep in mind what kind of operation we want to define for distributions. Also, there are some important theorems and lemmas for convolutions of functions that are useful to have at hand.

Theorem 2.3

Let $f \in L^1(\mathbb{R}^n)$ and, for $1 < p < +\infty$, $g \in L^p(\mathbb{R}^n)$. Then, $f*g$ is well-defined, $f*g \in L^p(\mathbb{R}^n)$ and:

$$\|f * g\|_{L^p(\mathbb{R}^n)} \leq \|f\|_{L^1(\mathbb{R}^n)} \|g\|_{L^p(\mathbb{R}^n)}$$

Lemma 2.2: Young's inequality

For $1 \leq\ p, q, r \leq +\infty$ such that:

$$\frac{1}{p} + \frac{1}{q} = 1$$

Then, for $f \in L^p(\mathbb{R}^n)$ and $g \in L^q(\mathbb{R}^n)$, this inequality holds:

$$\|f * g\|_{L^r(\mathbb{R}^n)} \leq \|f\|_{L^p(\mathbb{R}^n)} \|g\|_{L^q(\mathbb{R}^n)}$$